EART11300 Practical and Professional Skills

Week 8 – Unconformities

Introduction

In this class, we are going to focus on unconformities. As Bennison (2003) writes, "*an unconformity represents a period of time during which strata are not laid down*". In other words, an unconformity is a surface, or contact, between some overlying sediments and underlying older sequences of rocks, where there is evidence for a gap in time, which we call a 'hiatus'.

In simple cases, such a gap in time may be marked by an absence of deposition. More often, underlying rocks are uplifted, tilted, folded or eroded to varying degrees before deposition resumes. If the hiatus is particularly long, underlying beds may experience intrusion by igneous bodies such as dikes, or even undergo metamorphism. The ultimate result of any hiatus, usually best observed in the field, is that overlying strata rest on the eroded surface of an older set of beds (Bennison, 2003).

Unconformities are essential to understanding an area's full geological history, because the absence of rocks can be just as informative as the rocks which are present. The overall aims of today's class are to be able to:

- Define unconformities and categorize them according to their geometry, and relationships to units above and below;
- To identify unconformities in stratigraphic columns on published maps;
- To identify unconformities in map view, and draw simple cross sections and stratigraphic columns displaying these unconformities;
- [time permitting] To make sketch summaries of large areas of geological maps, highlighting such features as major faults, folds and unconformities

Categorizing Unconformities

Unconformities are categorized on the basis of:

(a) the shape of the unconformity surface and the relative orientation of the units on either side of it, or

(b) the relative orientation between the units above or below the unconformity surface and the unconformity surface itself.

Today we will focus mostly on (a). Those of you who have chosen to pursue Earth and Planetary sciences pathways next year may spend more time thinking about (b).

Before proceeding further, we need to define some more terms. An **angular unconformity** is one where the younger sediments lie upon the eroded surface of tilted or folded older rocks (Figure 1). The implication is that the tilting or folding of the older rocks occurred in the time-break represented by the unconformity. It is important to recognise that the dip and strike of the beds above the unconformity surface would be different from the dip and strike of the underlying beds.

In contrast, a **disconformity** is one where the overlying younger beds and underlying older beds have the same dip and strike (or dip direction). Broadly speaking, this means that there may have been

erosion of the older beds during the hiatus but that there was not significant tilting or folding. A special type of disconformity is a **paraconformity**, where the gap in strata is not very apparent, e.g. it may only be determined by dating of the rocks with fossils or isotopic methods. Disconformities and paraconformities could be together termed 'parallel unconformities', in contrast to more common angular unconformities.

Finally, a **nonconformity** describes where sediments are deposited onto an eroded surface of older igneous or metamorphic basement.

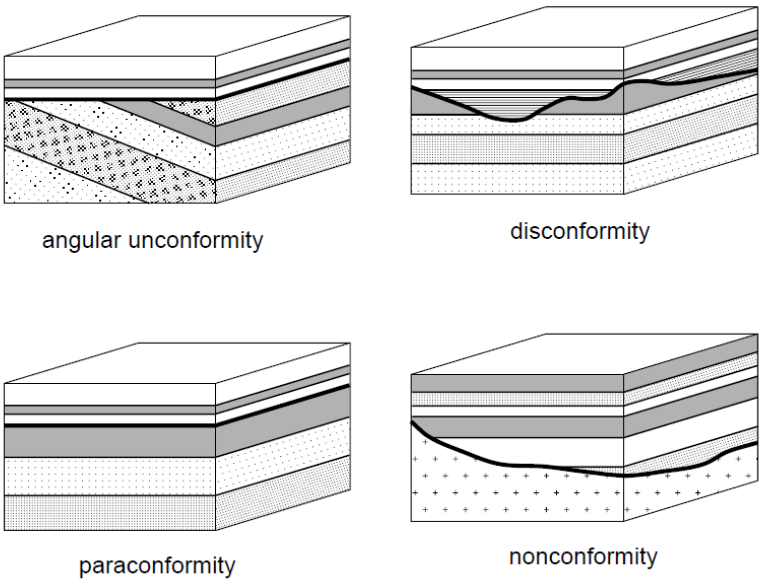


Figure 1. Sketch block diagrams showing the types of unconformity.

Exercise 1. Finding and Categorizing Unconformities

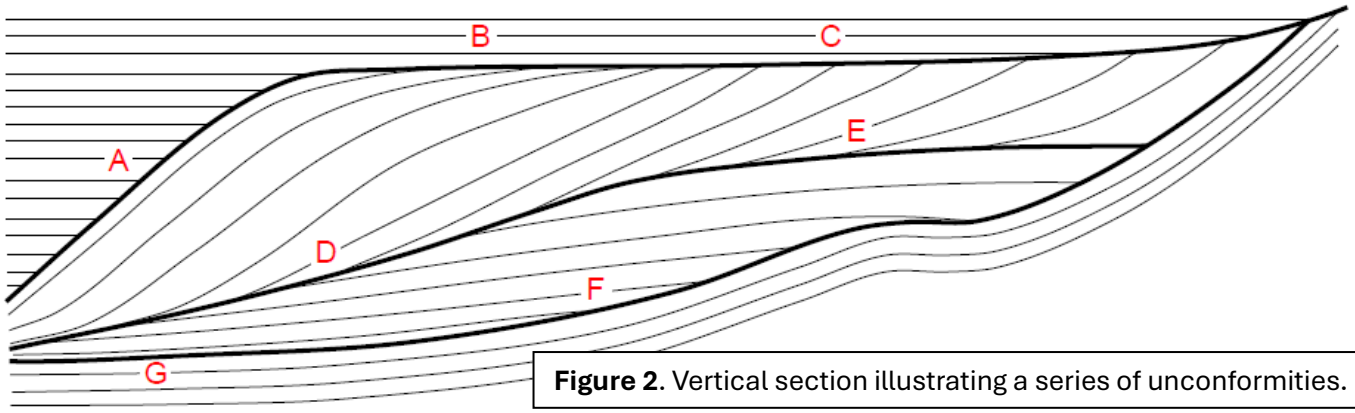


Figure 2. Vertical section illustrating a series of unconformities.

Study Figure 2, which is a vertical cross-section showing a series of geological strata, which are all right way up. The bold black lines represent unconformities. Considering each of the seven locations highlighted (A-G), state whether the unconformities at these locations are angular or parallel.

All are angular, except for G.

Next, consider the relative orientation of the underlying strata to the unconformity surface at each location. In most places, we are looking at 'concordance', which means the underlying strata and the unconformity surface are parallel. Can you find three locations where this is not the case and describe the geometry of the overlying beds relative to the unconformity?

B, C and D. For those who show interest, C and D is erosional truncation, where unconformity is an erosional surface which cuts into the underlying units; B represents toplap - truncation of initially inclined beds (*e.g.*, those formed on the foresets of deltas). Toplap typically results from non-deposition with minor erosion.

Recognizing Unconformities in Vertical Sections

Armed with the ability to recognise angular (or parallel) unconformities in cross-section, we will now turn to the British Geological Survey map of the Bristol area. There are some paper copies on the benches. You may also access it online, using the BGS map portal. There is a link in the Blackboard site.

This map is notable for a number of unconformities that we will study mostly using the **generalised vertical columns** ('stratigraphic columns') and **vertical cross-sections** provided in the margins of the map. Ask a GTA if you cannot find these.

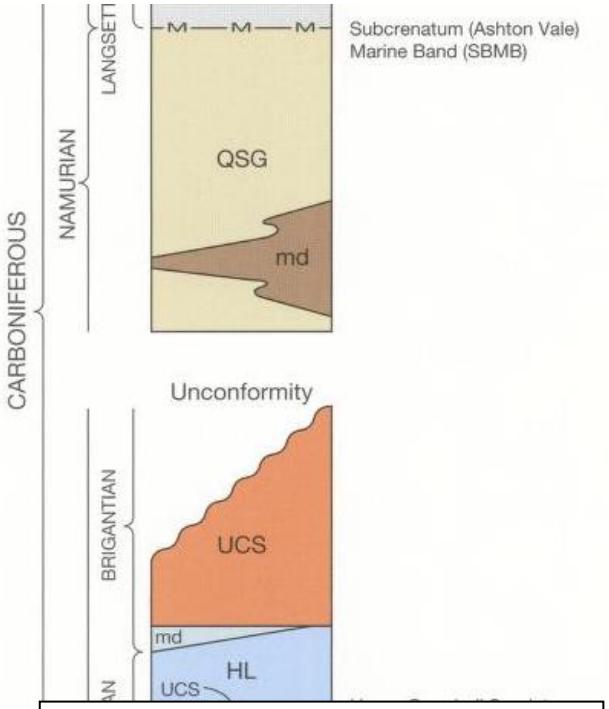


Figure 3. Excerpt from the generalised vertical section of the BGS Bristol map, showing how unconformities are presented, with squiggly line atop underlying unit.

Exercise 2. Unconformities on Generalised Vertical Columns

Find the generalised vertical columns on the Bristol sheet, and note the scale at which they are drawn.

1:6000 (Devonian-Carboniferous) and 1:2000 (Triassic- Jurassic)

Search through the columns until you find the Clifton Down Limestone Formation (CDL). What is the age of this formation, and what is the vertical thickness of this formation? [*Hint: there are multiple ways to determine the thickness*]

CDL – part of the Pembroke Limestone Group, Carboniferous Limestone, Lower Carboniferous, Visean (Mississippian). 153-277m

If I had a paper copy then I could measure the actual distance on the column that was labelled as CDL = 4.6cm – at a scale of 1:6000 = 27700cm = 277m. Alternatively, I could measure the thickness of the CDL formation from various places on the vertical cross section (thicknesses may vary) and convert them using the scale of the cross section. Sometimes, thicknesses are written on the Vertical Column itself. Can discuss why units appear thinner on cross sections (e.g. chopped by thrusts).

Now look for the unconformities on the vertical column (e.g. Figure 3). Starting with the oldest, write down the formations above and below the unconformity. Which geological periods do they belong to?

Above: Portishead Formation (within this formation is the Woodhill Bay Conglomerate member), Upper Old Red Sandstone, Devonian

Below: Black Nore Sandstone Formation, Lower Old Red Sandstone, Devonian

Now find the youngest unconformity on the section. What formations are above and below this unconformity? Which geological periods do they belong to? How long is the hiatus?

Above: Redcliffe Sandstone Formation, Mercia Mudstone Group, Triassic. Below: Publow Formation, Pennant Sandstone Group, Westphalian (Pennsylvanian), Carboniferous. The Permian geological period is absent. About 47 million years.

Exercise 3. Unconformities on Vertical Cross Sections

Now find the vertical cross-sections on the map. What scale are these drawn at?

1:50,000

Describe the dip of the beds in the cross-section, for units of (a) Carboniferous age and older, and (b) Triassic and Jurassic age. Explain your answer, giving evidence. [Hint: you may wish to look at the map itself as well as the section – find the line of section on the map and look for dip and strike (or dip direction markings close by]

The beds of Carboniferous age and older are dipping – I looked at the cross section

The Triassic and Jurassic age beds have generally shallower dip, are often nearly horizontal – I looked at the outcrop pattern on the map and their contacts follow the topographic contours meaning they are horizontal too.

Use the cross section to determine the nature of youngest unconformity. Choose one option from Figure 1 and justify your choice. Make a sketch of the key location/evidence from the cross-section (or take a screenshot and annotate it).

Angular unconformity. Look for 'cut-off points' where the contacts of the formations below the unconformity are truncated (cut off) by the line that represents the unconformity. This means that contacts of upper and lower beds are not parallel. For a normal stratigraphic contact, they would be.

Remember the unconformity is a special type of contact between formations

Exercise 3. Unconformities on Vertical Cross Sections

Now find the vertical cross-sections on the map. What scale are these drawn at? 1:50,000

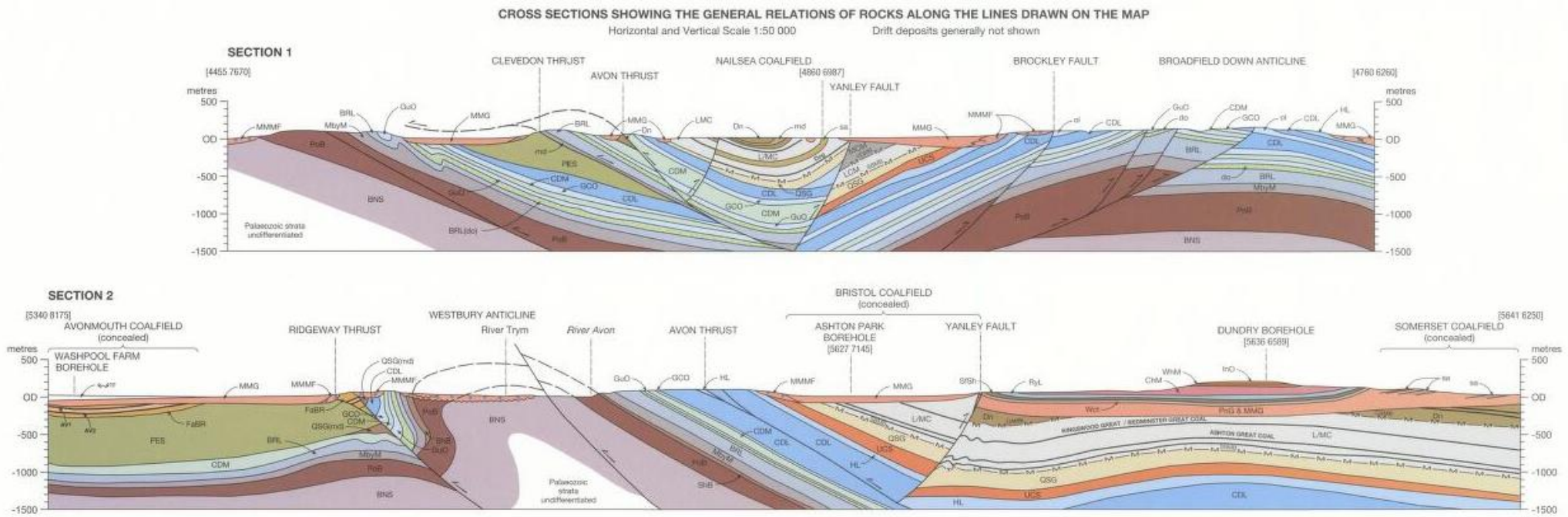
Describe the dip of the beds in the cross-section, for units of (a) Carboniferous age and older, and (b) Triassic and Jurassic age. Explain your answer, giving evidence. [Hint: you may wish to look at the map itself as well as the section – find the line of section on the map and look for dip and strike (or dip direction markings close by]

The beds of Carboniferous age and older are dipping – clearly visible on cross-sections. The Triassic and Jurassic age beds have generally shallower dip, are often nearly horizontal. Could also look at the outcrop pattern on the map: contacts follow the topographic contours meaning they are ~flat.

Use the cross section to determine the nature of youngest unconformity. Choose one option from Figure 1 and justify your choice. Make a sketch of the key location/evidence from the cross-section (or take a screenshot and annotate it).

Angular unconformity. Look for 'cut-off points' where the contacts of the formations below the unconformity are truncated (cut off) by the line that represents the unconformity. This means that contacts of upper and lower beds are not parallel. For a normal stratigraphic contact, they would be.

Upper section: look at MMG east of the Yanley Fault, truncating underlying UCS, QSG, LCM, MCM. Lower section: look at Yanley Fault here too.



Recognising Unconformities on the Map

You have now figured out how to spot unconformities in vertical sections, whether generalized vertical sections (squiggly line symbol) or cross-sections (truncations of lower beds). Now, we are going to figure out how to spot unconformities in map view, i.e. from above.

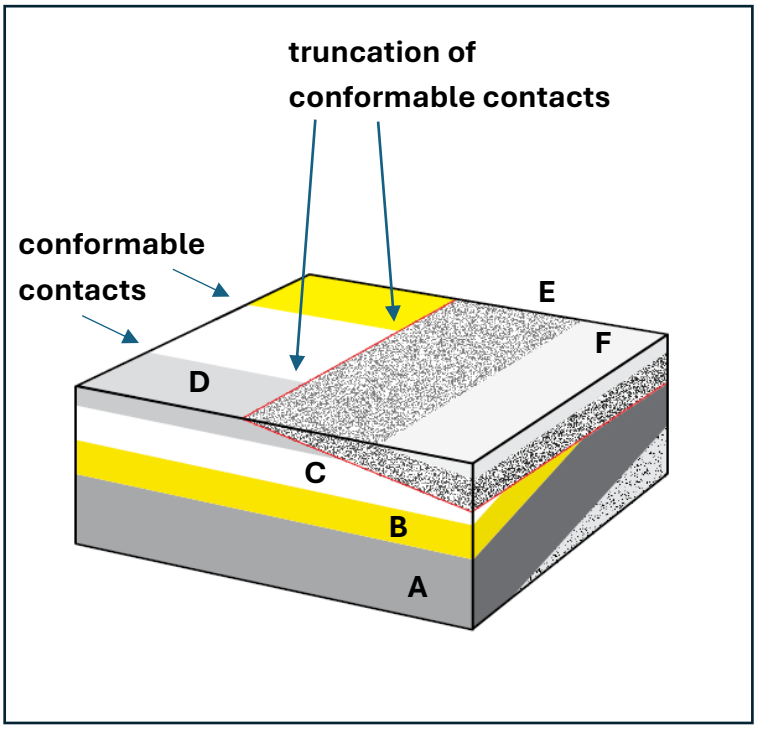


Figure 4. Block diagram from Houghton (2003) showing angular unconformity, with both vertical cross-sectional view and aerial/map view visible.

Look at Figure 4, which is a block diagram from Houghton's 'Introduction to Structural Geology', available on the Blackboard. An angular unconformity is shown. The vertical cross-section is similar to Figure 1. Look at the top surface now. This should give you an indication of how an angular unconformity will appear on the map. Contacts between conformably bedded older units (ie parallel lines between A, B, C, D) are truncated by the base of unit E. This is because, after the deposition of A-D they were somehow tilted, perhaps rotated, perhaps eroded, and unit D(+E) were deposited on top. Later erosion removed some of unit E and we can now see the angular unconformity below, from above. Ask a GTA if you are unsure about any of this!

Exercise 4. Unconformities in Map View

Now turn to Figure 5 on the next page. There are printed copies of this map available on the benches if you wish to work with pencil and paper.

(a) Figure 5 shows a sketched geological map with a series of different formations. Start by looking at each of these in turn. How many formations are there? Can you find the unconformity?

There are 6 formations, let's say A (cross hatching), B (dark grey), C (blank white), D (graph paper square grid), E (bold dots), F (little dots). The unconformity is below E. In the north centre of the map we can see truncation of the B-C contact and in the SW corner we can see truncation of C-D contact. In the SE corner we can see truncation of the B-C contact.

(b) Now, determine the orientation of the bedding both above and below the unconformity. You should specify the numbers in your answers both as dip/strike and dip/dip direction. [*Hint: you will need to use structure contours / stratum contours for this*]

150/13 SW, 13/240 below the unconformity. Use the base of B (dark grey).

090/1.5 S, 1.5/180 above the unconformity. Use the base of E (bold dots).

(c) Using your dip directions, now determine the sequence of units/formations.

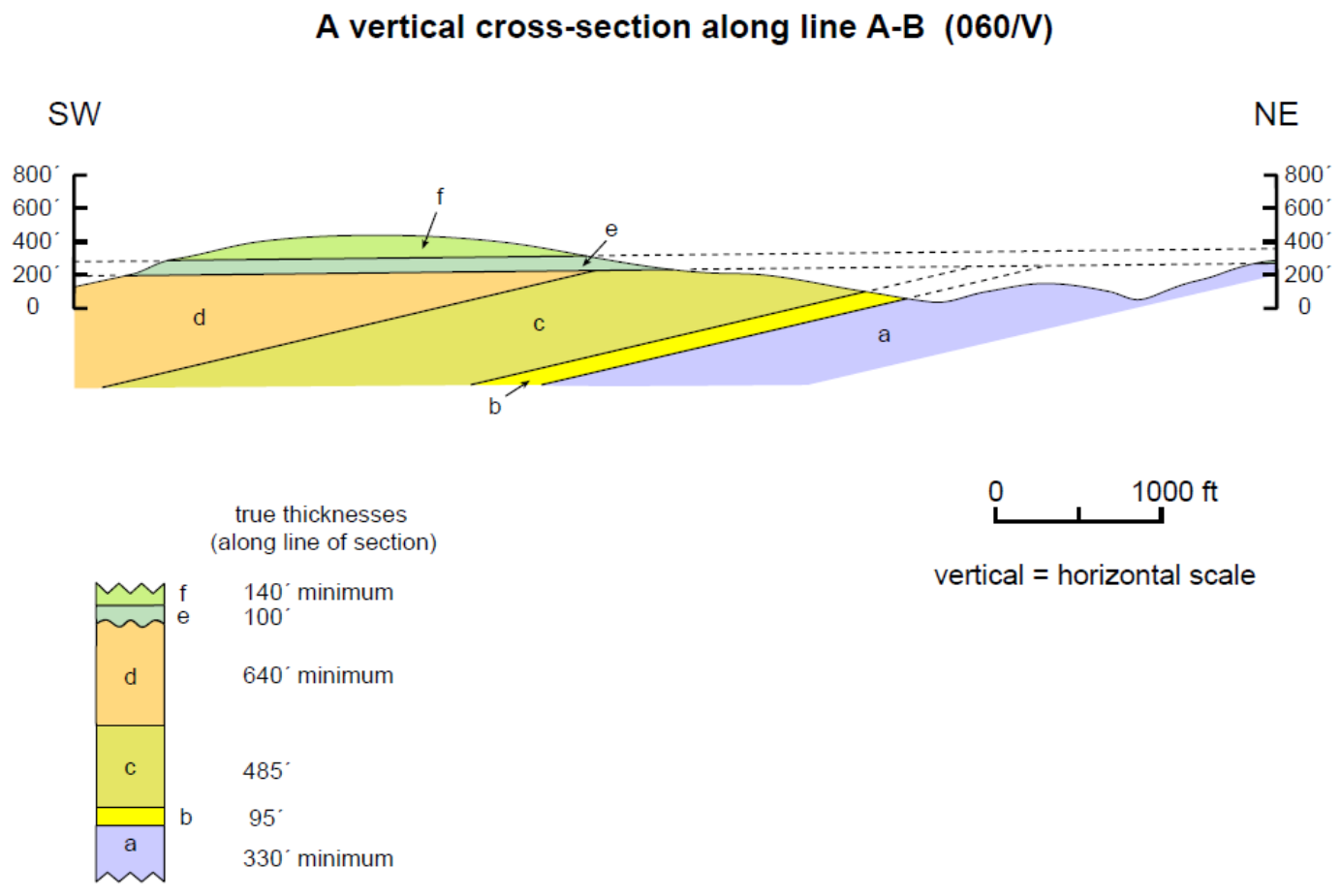
[old] A > B > C > D. Unconformity. E > F [young]

(d) Draw an accurate sketch cross-section along line A-B. You will find it much easier to do this using scrap paper and pen/pencil. You should remember to take photos of your sketches and copy these into your electronic notes.

[*Hint, if you are unsure how to draw a cross-section – we will demonstrate this in class: Start by marking the end points ('A' and 'B') at either end of a lightly drawn horizontal line. Next, move along this line from A towards B marking every time you cross a topographic contour. Using a suitable vertical scale, e.g. 1 cm = 200 feet (sorry, this is not metric…), now sketch in the topography along your cross section. Now, move along the line of section from A to B again, marking each time you cross a geological contact. Note the dip of the contact (using your dip and dip directions obtained above). Draw these contacts in below ground and consider projecting them a short distance above ground using dashed lines. Finally, use symbols or colour to highlight the different units*.]

(e) Lastly, draw a stratigraphic column / generalised vertical section **to scale**, with all formations labelled and either coloured or shaded with symbols to match the map. Determine the true thicknesses of all six units.

True thicknesses can be measured from the vertical cross-section, provided thickness measurements are made at 90 degrees to dip. If this can't be done, the width of the bed as exposed on the surface can be used, in combination with some basic trigonometry.



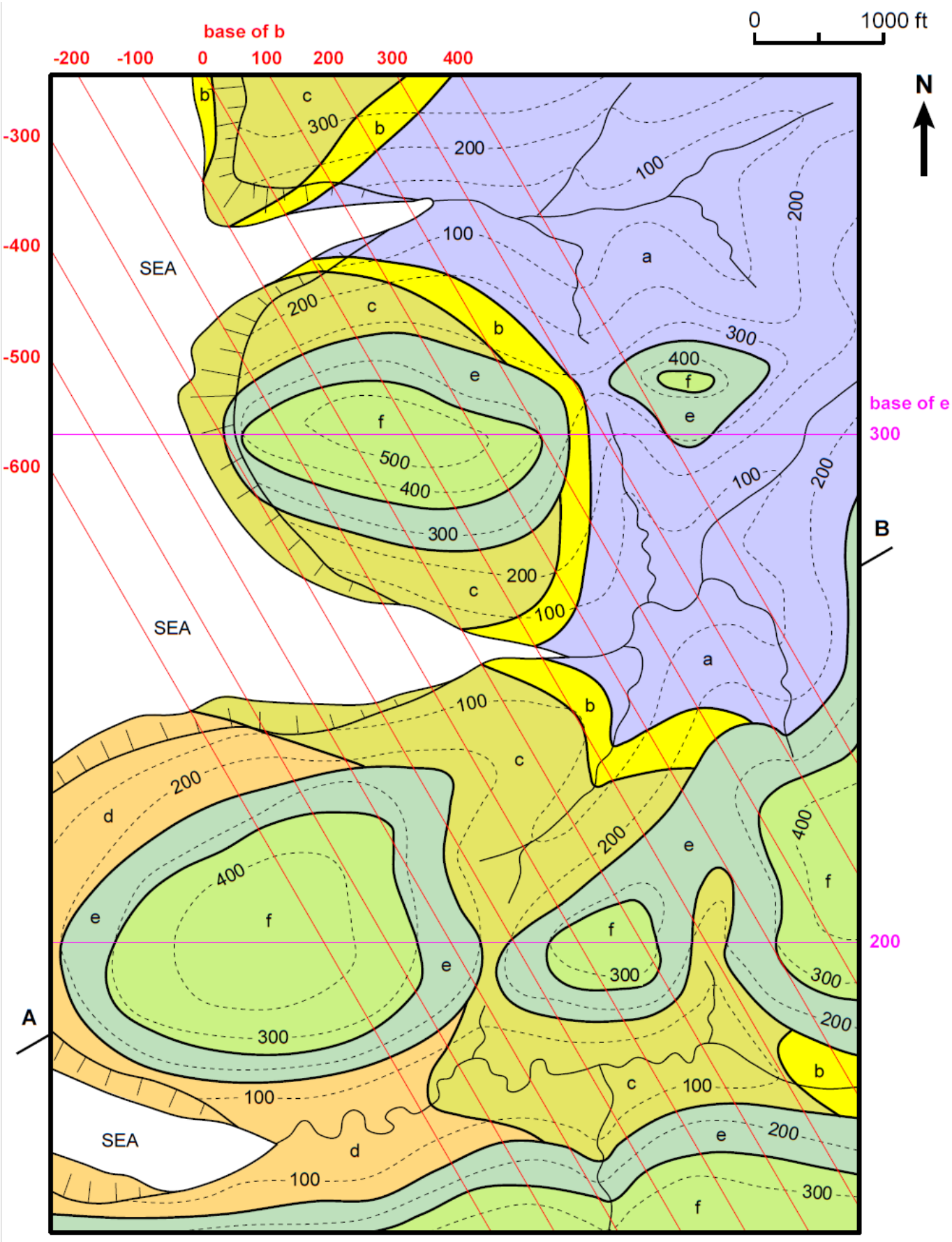


Figure 5. Sketch map illustrating an unconformity with a sequence of conformable units.

Summarising Geological Maps

The amount of data that are presented on geological maps can be very confusing at first glance. Sometimes we need to simplify what is presented to just show the main features. The drawing of cross sections is a very important part of interpreting the geology but another way is to draw a simplified sketch map.

The sketch map should show what you think are the important features but it does not contain the detail and complexity of the real map. You could use a sketch map to highlight particular features that were of interest to you e.g. where the unconformities or the positions of faults and folds are.

The last page of the handout lists some characteristics of both good and bad sketch maps. You should refer to these but note that today's sketch map is your first attempt, and you should simply try to do your best at summarising the geology, and not getting the map perfect.

Exercise 5. A Sketch Map of the BGS Shrewsbury sheet

Work in small groups and study the Shrewsbury map provided on the benches. Using the blank outlines provided in Figure 6 (and on the benches), you should attempt a summary sketch of the main map area.

- Focus on broad age divisions, i.e. "Permian-Triassic", "Carboniferous", "Pre-Cambrian and Lower Paleozoic". The last of these you will need to split into an upper and lower sequence (because there is an important unconformity to find).
- Add a generalized vertical section (without thicknesses). Show unconformities if you have found any.
- Show major faults and folds, if you can find them. Use the symbols you have learned (check these on the keys/legends of the maps if you can't remember how to mark faults and fold axes).
- Show some representative dips and dip directions.

This exercise will feel difficult, most likely. But talk to each other and look closely at the map and you will soon start to find the main features. Try to draw these accurately and annotate them appropriately.

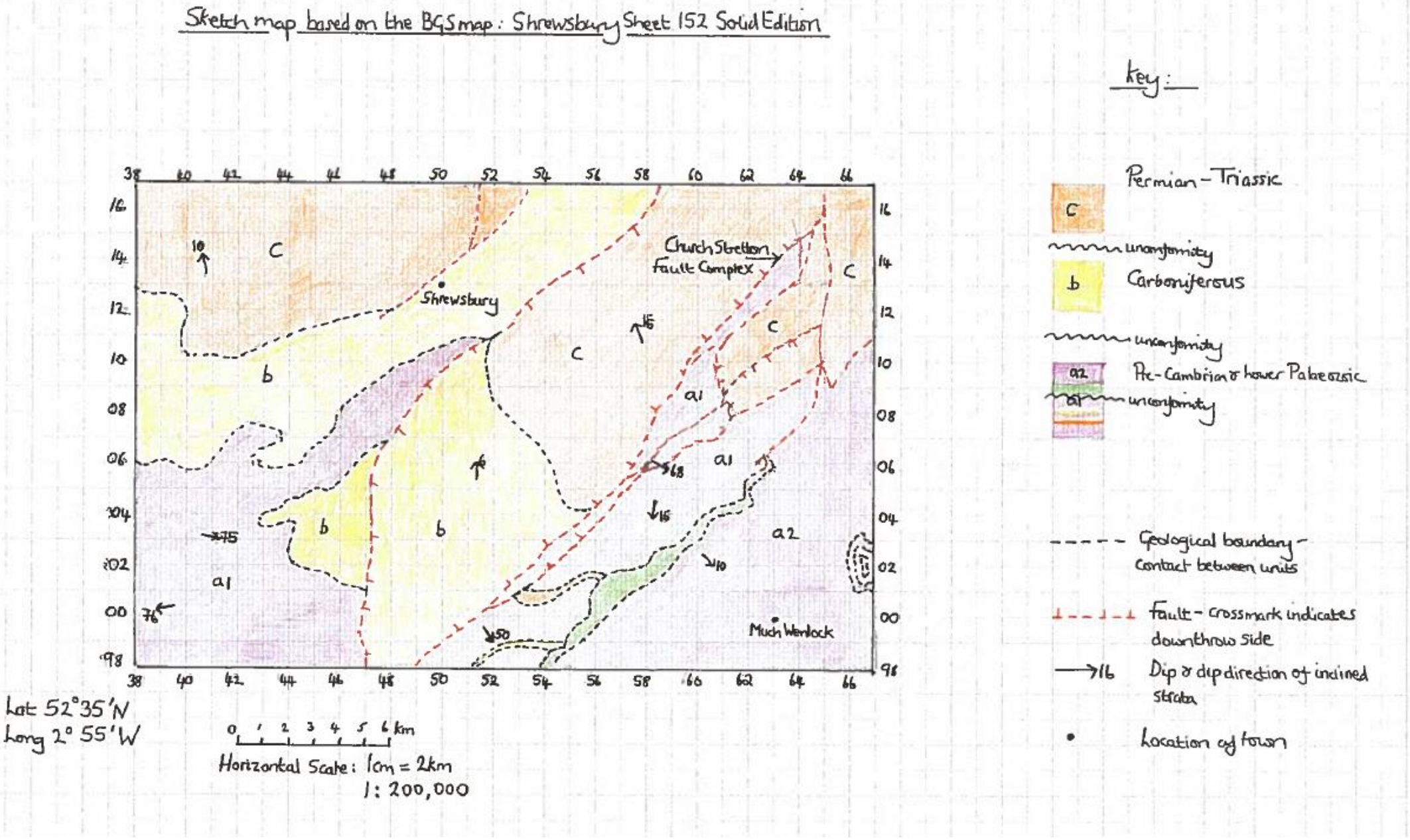


Figure 6. Sketch map of the BGS Shrewsbury sheet.

A good sketch map will show:

The main grid lines, a scale and a key along with a descriptive title explaining what the sketch map has been drawn to show. It will always be neat and easy to read.

It should use the same colours and letter keys as the original map

The outlines drawn accurately of the outcrop shapes of the important formations on the map. It may be useful to group formations together into larger units to keep the simplicity.

Some of the topographic features e.g. major river valleys and spot heights.

The main structural features such as the trend of major faults and the axial traces of folds

The strike and dip of formations if feasible

Annotations to explain features and complexities e.g. region of many parallel faults.

A bad sketch map:

Will have forgotten to number the grid lines, it will have no scale and no title to tell the viewer what the sketch map refers to.

It will not have a key and the colours and letter keys will show no relationship to the original map.

It will just be a detailed copy of the original map instead of a simplification or synthesis of the original.

It will just have copied the original formations and not grouped them into larger units.

A bad sketch map will not highlight the important features

It will give no indication of topography or strike and dip

It will have no annotations

It will be messy and hard to read.